

Create Superlative Images with OpenCV

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Open computing language, or OpenCV, as the name suggests, is an open source software that aims at computing efficiency and bettering real-time computer vision. An image processing software, OpenCV, helps us achieve whatever we want to do with images, right from understanding these to completely ripping these apart.

What can you do with OpenCV

Images can be of two kinds, namely, hand drawn or captured. In both cases, when we are dealing with images, there are two things we might want to do: understand the images completely and make these visually appealing.

OpenCV lets you do both, by providing a variety of functions and

features, which, when used correctly, give the desired result. With more than 2500 optimised library algorithms and new algorithms that can be written easily, it lets you work with images in an easy and comfortable manner.

Masking. A mask is a spatial filter. When we need to hide a certain region of an image, we can use a corresponding mask. It is like using a black-and-white filter on the image, wherein the black region blocks out the image, while the white region acts as a transparent sheet. An example can be seen in Fig 1.

Contrast and brightness. To enhance the image's appeal, we can play around with brightness and contrast features. Changing the contrast changes the difference between the selected region and its background, while brightness changes the appearance. We use the two to please our visual perception.

Smoothing, blurring, blending and sharpening. The anti-aliasing approach that finds usage right from image to audio processing is similar to smoothing and blurring the image.

Smoothing and

blurring help us nullify the difference between jagged edges and the rest of the image. These aid in generating background effects, shadows and even achieve region highlighting.

Blending involves mixing two layers by placing one on top of the other. Depending on the opacity of the two layers, the resulting effect can be hiding the bottom layer or dissolving the two or a transparent top layer. The reverse process; if we want to recover fine details of an image that is blur or low on resolution, we can sharpen the image.

Add noise. OpenCV lets us add a realistic angle to the image by corrupting it. This is like simulating the setup, taking into account noise from the detector, salt-and-pepper noise or data-transmission noise and also image-dependent noise.

Boost, identify and recognise.

We can follow eye movements in a video by tracking or measuring the point of gaze or the motion of the eye relative to the head. Removing red eyes in an image taken using flash is also a simple task. Boosting image classification by categorising images that can be used as a base in future. We can recognise scenery, as the tool compares the new image with previously-recorded ones.

The same goes with identifying or tagging people. Tracking objects and detecting motion in videos also take place using the same process.

OpenCV in real life

- Stitching street-view images together
- Detecting intrusions in surveillance videos in Israel
- Monitoring mine equipment in China
- Helping robots navigate and pick up objects at Willow Garage
- Detecting swimming-pool drowning accidents in Europe
- Running interactive art in Spain and New York
- Checking runways for debris in Turkey
- Inspecting labels on products in factories around the world
- Rapid face detection in Japan
- 2D and 3D feature toolkits egomotion estimation
- Human-computer interaction and mobile robotics
- Stereopsis stereo vision (depth perception from two cameras)



Fig. 1: Hiding an image behind a mask (Image courtesy: docs.coronalabs.com)